
The FASTag Bypass

How a one-man software shop gave highway contractors an unofficial lane around NHAI's toll system — an investigative analysis of the shadow-software case at ~100 highway plazas

CashlessConsumer Research

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1 The FASTag Bypass

1.1 How a one-man software shop gave highway contractors an unofficial lane around NHAI's toll system

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In one sentence: At roughly 100 Indian highway toll plazas, operators installed unauthorised “shadow” software on their own booth computers that printed real-looking receipts for cash taken from non-FASTag vehicles — while showing the national toll system almost none of it. The fraud ran for about two years across at least 14 states before a police sting, a parliamentary admission, a contractual-enforcement action that did not survive judicial review, and finally a federal money-laundering probe put the spotlight back on it.

1.2 Executive summary

What happened. For roughly two years (March 2023 to January 2025), operators at a toll plaza on National Highway 35 in Mirzapur, Uttar Pradesh ran a piece of unauthorised software — known in the case as “Mobdata” and “Any” — on the plaza’s own booth computers. The software let booth operators record payments from vehicles without a valid FASTag sticker, collect cash from those drivers, and then make almost all of those transactions vanish from the official accounting system. Drivers paid. The government did not get its user fee. Only about 5% of the shadow transactions ever entered the official record.

How big. What UP Police’s Special Task Force found at one plaza in January 2025 turned out to be a template. Police and NHAI traced the same pattern to about 42 plazas within weeks; by September 2026, when the Enforcement Directorate filed a money-laundering case, the number was about 100 plazas across more than 14 states, with searches at 14 premises in 6 states. The STF’s working estimate at the original plaza alone was ₹45,000 of skimmed cash per day; police put total diversions across the identified network at over ₹120 crore. NHAI encashed over ₹100 crore of bank guarantees from defaulting agencies in March 2025 — then saw the debarment orders set aside by the Delhi High Court in July 2026, while contract terminations and forfeitures were left to contractual remedies for some agencies.

The policy backdrop. The state projected digitisation as the integrity fix for tolling: FASTag mandatory from November 2024, penetration above 98 percent, double fees for cash payers from November 2025. This report documents the gap between that projection and the on-ground state of the systems - two years of unauthorised software that no official control detected, and a judicial record that, in July 2026, diverged sharply from the official account of the same events.

Why it matters. This was not a hack. Nobody broke the FASTag encryption or spoofed a tag. The “bypass” was human trust plus unaccountable software on machines that only the operator’s own staff could touch — a hole that every audit report and reconciliation system in the chain was structurally blind to, because the shadow software kept the *appearance* of a normal system: a steady trickle of “exempted” and “violation” receipts, bank remittances that matched the (falsified) books, and drivers walking away with legitimate-looking receipts.

What this report does. It explains the toll-plaza system from scratch for readers who have never thought about what happens at a toll booth; reconstructs the bypass from police chargesheets and press records; measures the footprint against official open data (the IHMCL plaza registry and NHAI’s own Toll Information System); follows the accountability arc through three acts — crackdown, court reversal, and federal money-laundering probe; and critiques the three procurement documents NHAI has published since, which quietly concede that lane-level cash is the system’s unmonitored blind spot.

1.3 Why you should read this

If you have ever driven on an Indian national highway, this case is about a system you paid into — and about the small fraction of payments that still happen outside the digital trail. It explains:

1. **How tolling actually works** — the hardware, the software, and the little-known manual side-door that still existed at every plaza.
2. **How the bypass worked** — in plain language, reconstructed from police and enforcement-directorate findings.
3. **Why the audits missed it** — a systems failure worth understanding, because it will happen again in other digital-public-infrastructure systems.
4. **What happened to accountability** — including the part where the High Court undid the regulator's punishment, and what happened next.
5. **What the open data shows** — we matched the scam plazas against the government's own plaza registry, and the results are uncomfortable.
6. **What changes now** — and what risks are simply migrating to the next system.

This report is written for a reader with **zero background** in tolling systems or this case. Every technical claim is sourced; every unverified claim is labelled as such. A full evidence archive underlies this report (see Appendix C).

2 Part I — The system, before the scam

2.1 1. A toll plaza, disassembled: the lane, the reader, the booth computer, and the money

India’s national highways (the “NH” network) are largely funded through **user fee** — the toll you pay at a plaza. There were, as of the government’s own registry from August 2025, **more than 1,100 fee plazas** across 24 states and union territories (we count them in Part IV).

A toll plaza is not run by the government. It is run by a **fee-collection agency** — a private contractor engaged by the concessionaire (the company that built or operates the highway section) under the overall supervision of the **National Highways Authority of India (NHAI)**, a statutory body under the Ministry of Road Transport and Highways (MoRTH). A separate NHAI joint venture, the **Indian Highways Management Company Limited (IHMCL)**, runs the technology backbone of electronic tolling.

At the booth, the stack looks like this:

Layer	What it is	Who controls it
Lane hardware	RFID (FASTag) reader, barrier arm, vehicle-classification sensors, CCTV/ANPR camera, thermal receipt printer	The concessionaire’s vendor
Toll Management System (TMS)	The lane software: reads the tag, classifies the vehicle, computes the fee, records the transaction, prints the receipt	The fee-collection agency’s technology vendor
Back office	Batch settlement of the day’s transactions to the acquiring bank, then to IHMCL/NPCI’s NETC clearing system	Agency + bank
Oversight	NHAI Regional Offices inspect plazas; IHMCL monitors NETC transactions; ministry committees review	Government

The critical sentence in that table is the third row of the first column: **the TMS lane software is procured and controlled by the private collection agency, not by NHAI**. Hold that thought.

2.2 2. FASTag: how a windshield sticker came to carry 98 percent of India's highway payments

Since February 2021, paying toll on national highways by cash has been formally the exception, not the rule. Vehicles carry a **FASTag** — an RFID sticker linked to a prepaid wallet or bank account. As you approach the plaza:

- The reader identifies the tag, the lane software computes the fee for your vehicle class, the barrier lifts, your account is debited, and the transaction flows into the **National Electronic Toll Collection (NETC)** ecosystem operated by NPCI (the same organisation behind UPI).
- Radio-frequency tags fail sometimes — dead battery in the reader, damaged sticker, insufficient balance, or a tag listed as blacklisted. For exactly these vehicles, the barrier stays down and the booth operator keys the vehicle class manually, charges a **double fee** (the “2× penalty” for travelling without a valid tag), prints a receipt and accepts **cash**.

In ministerial statements to Parliament, roughly **98% of transactions** were electronic; the non-FASTag residue was about **2% of vehicle volume**¹. Two per cent sounds trivial. It is not. Across a national network handling hundreds of millions of trips, the non-FASTag lane is a constant, cash-heavy, operator-controlled stream — and, as this case shows, it was being **skimmed at industrial scale**.

2.3 3. The manual side-door: how vehicles without a valid tag pay, and why that lane was never really watched

Here is the design weakness at the heart of this whole story, and it was visible in official documents years before the scam surfaced:

- The operator's keystroke in the TMS decides what the national system believes happened at the lane. If the operator marks a paying vehicle as “**exempted**” (exempt category) or “**violation**” (jumped the barrier, or could not be charged), no toll is recorded as due.
- There was **no independent physical count** at the lane — no reconciling of barrier events or sensor counts against keyed transactions.
- The agency's own technology vendor controlled the lane software that produced the receipt and the record.
- A program to monitor whether agencies actually **remit** what they collect (“Toll Remittance Alert System”) was only being tendered in April 2023 — *after* the fraud had already begun².

In short: the national system trusted the lane, and the lane belonged to the contractor.

¹Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

²IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

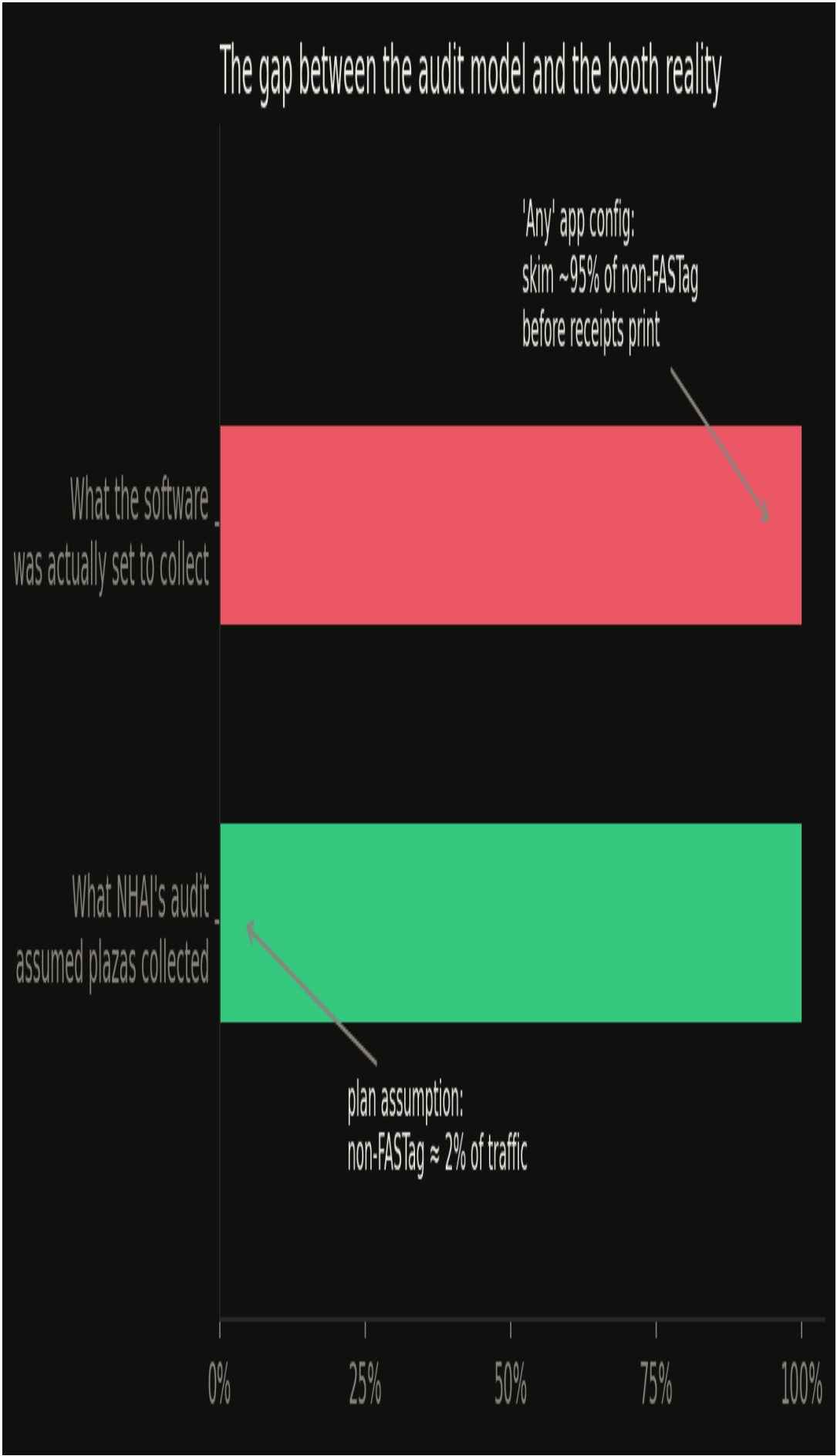


Figure 1: The gap between the audit model and the booth reality. What the official record shows versus what drivers actually paid at a scam plaza: roughly one in twenty shadow transactions entered the TMS, each shadowed by a plausible-looking printed receipt. Chart: CashlessConsumer reconstruction from STF/ED case reporting.

3 Part II — The scam

3.1 4. January 2025: a checkpoint sting at one Mirzapur plaza blows open a network of forty-two

On 22 January 2025, Uttar Pradesh police's Special Task Force (STF) registered FIR No. 0017/2025 at Lal-ganj police station, Mirzapur district, and moved on the **Atraila Shiv Gulam toll plaza** on the Varanasi–Hanumana section (old NH-7, now NH-135) — a plaza on the pilgrimage-and-coal corridor southeast of Mirzapur³.

Within two days the STF had:

- Arrested **Alok Kumar Singh**, 35, an MCA (computer-applications) graduate from Jaunpur district who had briefly worked at a toll plaza in 2021. He was picked up near Babatpur airport, Varanasi⁴.
- Detained the plaza's manager, **Rajeev "Raju" Mishra** of Prayagraj, and an operator, **Manish Mishra** of Sidhi (MP)⁵.
- Seized two laptops, a printer, five mobile phones, ₹19,580 in cash and a vehicle⁶.

What the STF described was not a cash-skimming trick by a rogue operator. It was a **software product**, allegedly built by Singh, that had been **installed on the plaza's own booth computers** and operated in parallel with the official TMS⁷.

3.2 5. Anatomy of the bypass: 'Mobdata' and 'Any', the shadow software that stood between the booth and the books

Pulling together STF briefings, accused interrogations and later ED disclosures, the machine had these parts:

1. The trigger. A vehicle arrives without a valid FASTag. Barrier stays down. Under the rules, the operator must charge double the fee, in cash, and the transaction must be keyed into the official TMS (the 2x rule that took effect 15 Nov 2025)⁸.

2. The shadow app. On the booth computer, alongside the TMS, ran an unauthorised application (referred to in ED statements by the names '**Mobdata**' and '**Any**' — see the caveat in §11). It computed the **same double fee** from the public gazette fee schedules, so the receipt it printed was arithmetically indistinguishable from a genuine one⁹.

³<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁴<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁵<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁶OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁷Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

⁸National Highways Fee (Third Amendment) Rules, 2025 - non-FASTag vehicles paying cash charged double fee, effective 15 Nov 2025. PIB, 4 Oct 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2174761>

⁹<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

- 3. The forged receipt.** The app used the **same thermal printer path** as the TMS. The driver got a plausible receipt showing the toll and the 2× penalty. Nothing the driver could see was wrong¹⁰.
- 4. The vanish.** Only about **5%** of these non-FASTag transactions were keyed into the official TMS — enough to keep the “exempted/violation” categories looking ordinary. The other 95% existed only inside the shadow app¹¹.
- 5. The monitoring layer.** Plaza-level collections were pushed to a **central monitoring system** — ED describes separate monitoring portals; STF found Singh’s laptop had direct access to plaza data across plazas¹²¹³.
- 6. The split.** Cash stayed at the plaza level; Singh’s share moved through at least five bank accounts and digital wallets, according to ED findings reported in September 2026¹⁴.

3.2.1 Why the scam was invisible from above: how a falsified system still looked perfectly normal

Do the arithmetic the way an auditor would. Non-FASTag traffic is about 2% of volume. Skimming 95% of it removes **under 2% of gross revenue** — a rounding error in aggregate charts, easily masked by genuine monthly variance. And every audit trail NHAi could query — the TMS database, the NETC settlement records, the bank remittances — was fed by the very lane software the shadow app sat beside. **The fraud never wrote to the system that was being audited.**

That is why, when NHAi’s technical team examined computers at plazas in Gorakhpur after the STF raids, it reportedly found “no significant evidence”¹⁵. Auditing the books cannot catch a second set of books.

3.3 6. The footprint: from one plaza on NH-35 to about a hundred plazas across fourteen states

Date	Who says	Count
23 Jan 2025	Day-1 interrogation claims	“up to 200 plazas” (confession-based, never confirmed) ¹⁶

¹⁰<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

¹¹OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹²OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹³<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

¹⁴<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

¹⁵Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

Date	Who says	Count
23 Jan 2025	STF verified list	42 plazas in 14 states, operators named ¹⁷
13 Feb 2025	STF	150–200 plazas “under examination”; notices to managers in 6 districts ¹⁸
3 Sep 2026	ED (device forensics)	~100 plazas where the apps were used ¹⁹

¹⁶Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

¹⁷<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

¹⁸<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

¹⁹The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

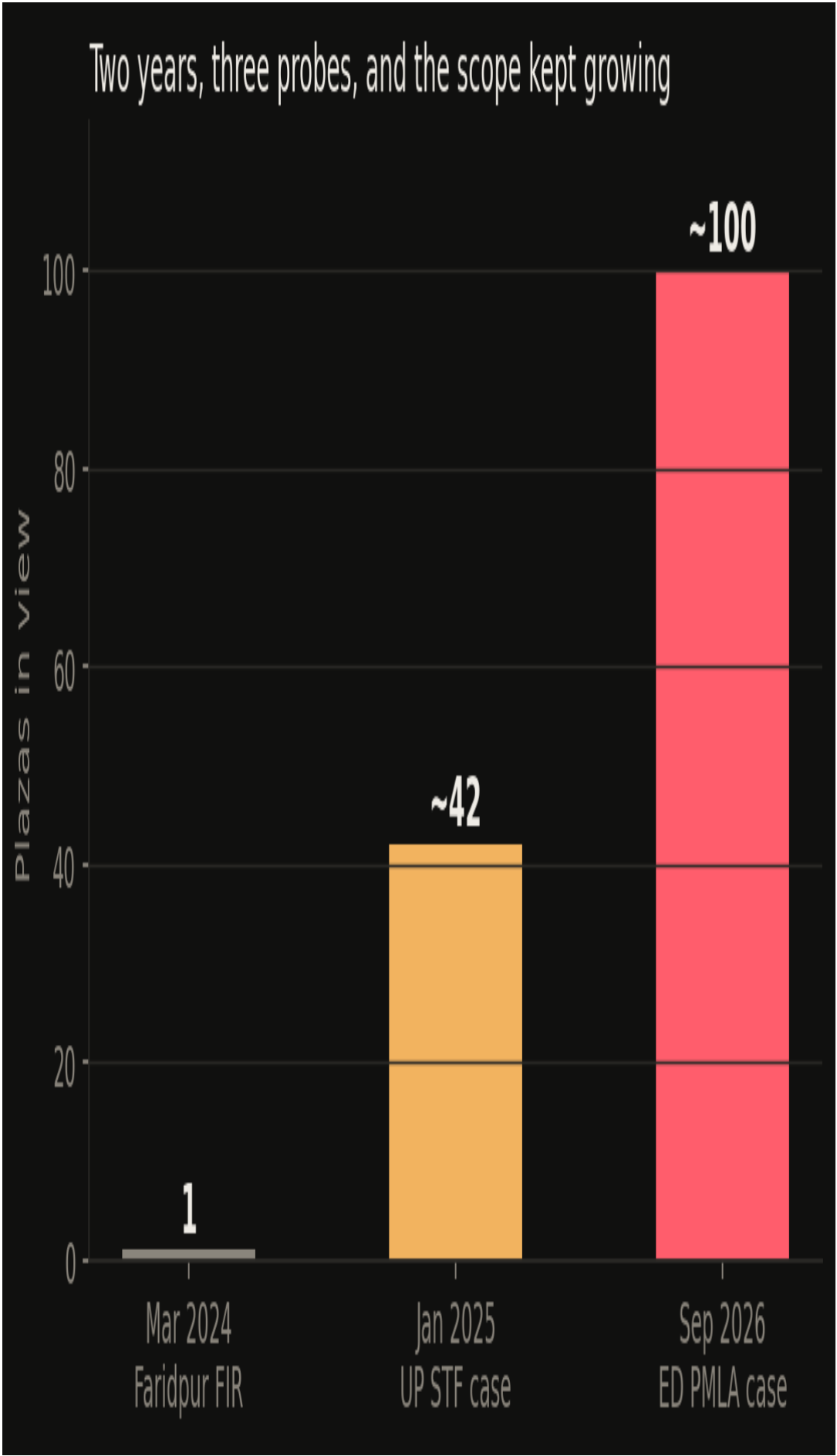


Figure 2: How the number grew. What police found at one plaza in January 2025 was a deployable template; each official acknowledgement expanded the count. The September 2026 ED figure (~ 100 plazas) is an admission count, not a ceiling.

The verified pattern: 42 was the *arrest perimeter*, not the ceiling. ED's forensic examination of seized devices in 2026 nearly tripled the confirmed footprint, and pointed to plazas in **UP, Rajasthan, Madhya Pradesh, Jammu & Kashmir, West Bengal and the Delhi-NCR** among the searched premises²⁰²¹.

²⁰<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²¹<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

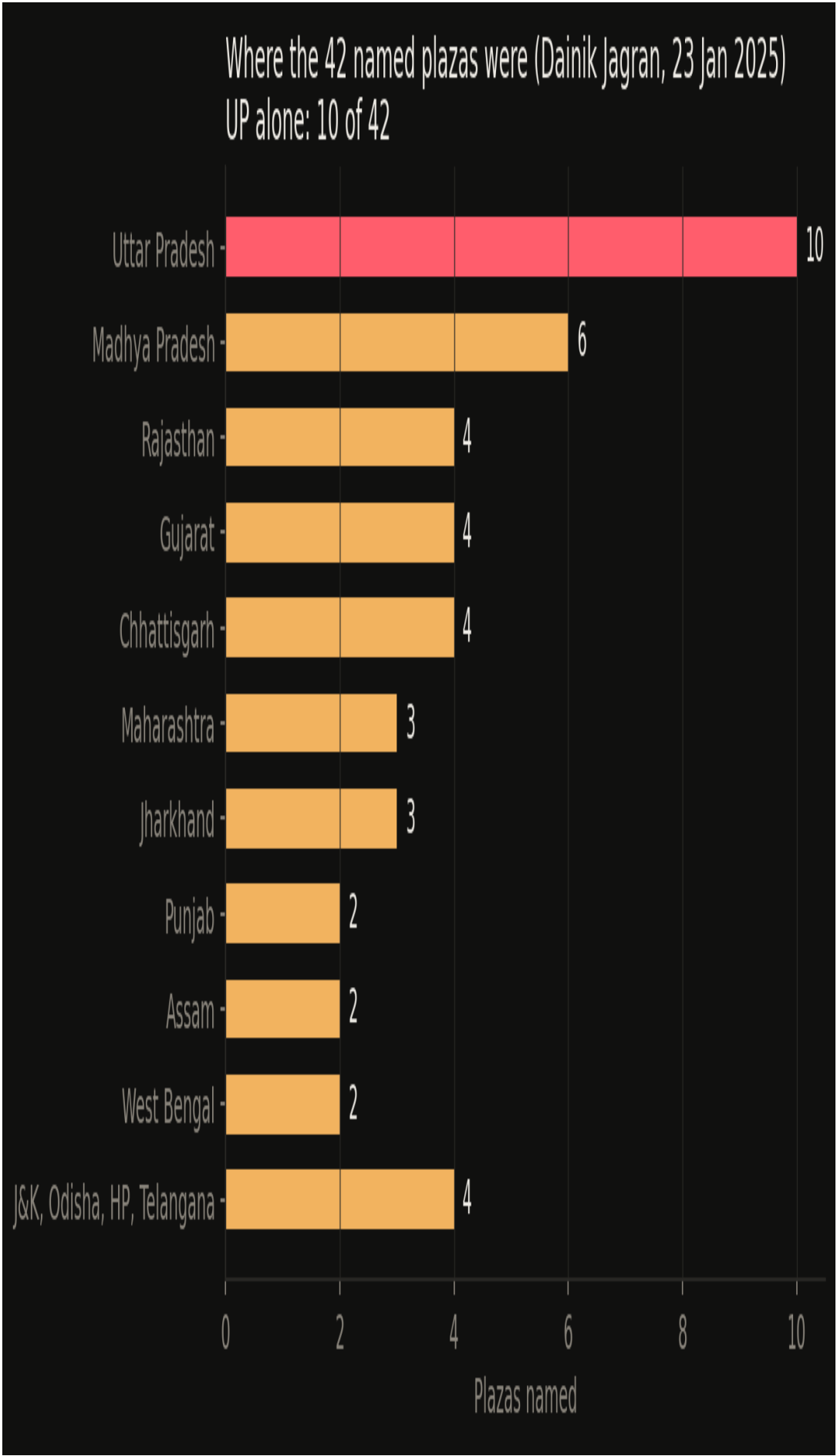


Figure 3: Where the 42 verified plazas were. State-wise split of the STF’s January 2025 verified list, as reported by Dainik Jagran. UP, MP and Rajasthan alone account for 20 of the 42. Source: press release by CashlessConsumer Research, compile in the evidence archive.

3.4 7. Following the money: ₹45,000 a day at one plaza, ₹120 crore across the network, ₹100 crore in guarantees

- **Per plaza:** STF's working estimate at Atraila was roughly **₹45,000 per day** of skimmed cash²². Over the ~730 days the fraud reportedly ran, that is about **₹3.3 crore from a single plaza** — a figure Dainik Bhaskar separately reported as the “3.28 crore embezzled at one location”²³.
- **Network scale:** the STF put total diversion at **₹120 crore+** across the 42 plazas²⁴. Early day-1 media claims reached ₹750 crore — never substantiated, treat with caution.
- **ED's 2026 language:** receipts for unauthorised collections “running into crores” across ~100 plazas; ₹1.03 crore cash seized in the September 2026 searches and **eight bank accounts** frozen²⁵.

3.5 8. The humans: an MCA graduate, a plaza manager, a booth-operator network, and the agencies that hired them

Actor	Role	Status
Alok Kumar Singh	Alleged developer of the shadow software; MCA, ex-toll-plaza worker	Arrested Jan 2025
Rajeev “Raju” Mishra	Manager, Shivgulam plaza (Prayagraj native)	Arrested Jan 2025
Manish Mishra	Booth operator (Sidhi, MP)	Arrested Jan 2025
Saaban Lal Kumhavat	Fourth arrest	Feb 2025 ²⁶
“Sawant” and “Sukhantu”	Alleged system managers for the 42-plaza operation	Named in STF briefing; status unclear ²⁷
“Veltek” (per Dainik Jagran)	Collection agency named for Atraila plaza	Identity unresolved; not among the debarred 14 ²⁸

²²<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

²³Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

²⁴<https://www.thedailyjagran.com/india/toll-collection-fraud-nhai-bans-14-agencies-for-using-fake-software-to-collect-toll-illegally-rs-100-cr-security-confiscated-10225098>

²⁵<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²⁶<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

²⁷OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

²⁸<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

Publishing caution: several names above come from single-outlet or briefing-based reporting. We label them as alleged, and we do not name private individuals beyond what multiple independent reports establish.

4 Part III — Accountability, in three acts

4.1 9. Act one (March 2025): NHAI's enforcement - fourteen agencies debarred, ₹100 crore of securities encashed

The STF case triggered the fastest nationwide contractual crackdown NHAI had executed:

- In Parliament (20–21 March 2025), Minister Nitin Gadkari confirmed the bypass: vehicles without valid tags were being recorded as “exempted or violation” with receipts never entering the system — while insisting there was **no breach of the electronic tolling system itself**²⁹. Announcements followed: AI-based audit cameras and ANPR pilots.
- NHAI **debarred 14 user-fee collection agencies** for one to two years, **encashed performance securities reported at over ₹100 crore**, and levied penalties exceeding ₹2 crore³⁰³¹.
- Among the debarred: AK Construction (linked to nine plazas on the scam list), RK Jain Infra Projects, Vanshika Construction, Innovision Ltd and others³². The Atraila agency was separately terminated with its bank guarantee encashed.

4.2 10. Act two (July 2026): the Delhi High Court tests NHAI's case against its record

On 3 July 2026, the Delhi High Court (Justice Sachin Datta) disposed of a batch of writ petitions - *T. Suryanarayana Reddy v. NHAI*, W.P.(C) 10788/2025 and connected matters - filed by toll-collection agencies against NHAI's February 2025 action: debarment, contract terminations and performance-guarantee forfeiture³³³⁴. The lead judgment is *T. Suryanarayana Reddy* (full text archived: Appendix C). The outcome was a split, and the reasoning matters as much as the result.

The outcome. All debarment orders were set aside³⁵. The court declined to interfere with the contract terminations and performance-guarantee forfeitures of five petitioners (W.P.(C) 11106, 11172, 11684, 12044 and 12091 of 2025), leaving them to pursue contractual and civil remedies (paras 61-62), and expressly recorded that it expressed no opinion on the merits of those claims (para 62). The reputational penalty - blacklisting - fell; the financial consequences partially survived.

NHAI had already tasted judicial pushback months earlier: in September 2025, the same court had stayed the debarment of Innovision Ltd after its show-cause notice cited a plaza the company did not operate³⁶. The July 2026 batch judgment generalised that first crack into a systematic finding.

²⁹ Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

³⁰ <https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³¹ <http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

³² <https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³³ <https://indialegallive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

³⁴ <https://indiankanoon.org/doc/99533796/> (lead matter, W.P.(C) 10788/2025, decided 03-07-2026). Full archived copies in Documents/research/FASTagScam/legal/.

³⁵ <https://indiankanoon.org/doc/99533796/> (lead matter, W.P.(C) 10788/2025, decided 03-07-2026). Full archived copies in Documents/research/FASTagScam/legal/.

³⁶ <https://indiankanoon.org/doc/159332680/> (*M/s Innovision Limited v. NHAI*, interim order, 12 September 2025).

Why the debarments failed. The court's findings track the record as it stood in February 2025, and each finding is a finding about evidence, not about the scam's underlying truth:

- **No forensic examination.** The devices seized in the 14 January 2025 STF raid - laptops, phones, the booth computers - were never sent to the Forensic Science Laboratory before NHAI acted. The core allegation (unauthorised software on shared booth computers) carried a built-in attribution problem the record never resolved: several operators shared one booth computer, so presence of the app did not establish which agency, if any, had knowledge of it.
- **The data inference was “fraught with difficulties”.** NHAI's case rested on revenue-pattern inference: cash collections surged after the raid, therefore pilferage must have occurred before it. The court found this probabilistic reasoning insufficient for “objective satisfaction” (paras 46-48). Petitioners produced concrete counter-examples: at Baleni plaza the post-raid increase was ₹1,000-1,200 on cash volumes below 1 percent of total revenue - a large *percentage* on a tiny base; at Rohisa, the surge partly pre-dated the raid itself; and one petitioner's exemption and cash ratios moved *against* the direction the alleged modus operandi predicts (para 47).
- **Unexcluded external factors.** The period coincided with the Maha Kumbh Mela and pilgrimage-season traffic; the court found no cogent basis to rule out these explanations for cash surges (para 49).
- **The legal standard.** Debarment is civil death and demands “strong, independent and cogent material” (para 54). The orders, in the court's words, rested on “a ‘theory’ rather than any ‘objective’ facts” - an attempt “to reason backwards ... to convert ‘suspicion’ into ‘guilt’ in the absence of a single piece of independent evidence” - and allegations of fraud must meet a higher threshold of proof, per *Union of India v. Chaturbhai M. Patel* (para 55). The debarment orders could not stand.

The press release and the record. NHAI's 11 February 2025 press release had presented the February action as established fact: fourteen agencies, violations “found”, ₹100 crore in securities. The High Court's record review found the underlying file contained no forensic examination, contested statistical inference and unexamined alternative explanations. The episode is a case study in the gap this report tracks: the public narrative asserted more than the administrative record could support, and it took judicial review - seven months later - to expose the difference. Consumers reading the February headlines had no way to know the case was a theory awaiting proof.

What the judgment did not decide. The court set aside the process, not the underlying allegation. Para 62 preserves NHAI's freedom to establish its case through proper proceedings, and the ED's PMLA investigation (Act three) proceeds on an independent statutory footing, building the forensic record the February orders lacked. A reader should hold both facts: the contractual accountability layer as constructed did not survive scrutiny, and the question of what actually happened at the plazas remains open before the criminal-forensic track.

4.3 11. Act three (September 2026): the Enforcement Directorate arrives, and the case becomes about laundering

Which brings us to the news that started this investigation: on **2 September 2026**, the **Enforcement Directorate (ED)** — the federal financial-crime agency — conducted searches at **14 premises across six states** under the Prevention of Money Laundering Act (PMLA), seized ₹1.03 crore in cash and froze eight bank accounts³⁷. On 3 September, the ED disclosed that its forensic examination of devices pointed to the unauthorised apps and monitoring portals at **about 100 plazas**³⁸³⁹.

Read in sequence, the ED action is the state's **second attempt at accountability** — this time using criminal-forensic machinery (device images, banking trails, portal data) rather than contractual penalties, after the contractual-penalty route did not survive judicial scrutiny (section 10).

On the names ‘Mobdata’ and ‘Any’: these app names appear only in ED-attributed wire copy. We searched app stores, code-hosting platforms, and certificate-transparency logs; **no public trace of any product by these names exists**. They are likely internal codenames visible on the seized devices and portals. We treat them as attributed claims, not established product names — and we note that “MobData” is also the name of an unrelated Chinese analytics SDK, which should not be conflated.

³⁷<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

³⁸The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

³⁹<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

5 Part IV — What the open data shows

This is the part you cannot get from news reports, so we built it.

5.1 12. Counting every plaza: what the official registry says (and lets anyone check)

IHMCL publishes the official list of national-highway fee plazas as a PDF⁴⁰. We parsed it (methodology in Appendix A) and extracted **1,133 plazas**, each with its NETC plaza code, name, state and district.

Where the plazas are:

State	Plazas
Rajasthan	163
Uttar Pradesh	135
Maharashtra	101
Madhya Pradesh	89
Tamil Nadu	81
Andhra Pradesh	79
Haryana	78
Karnataka	62
Gujarat	56
Punjab	41
Bihar	41
Telangana	36
Odisha	32
West Bengal	29
Chhattisgarh	23
Jharkhand	22
Delhi	13
Assam	10
Kerala	8
Uttarakhand	8
Jammu & Kashmir	7 (+1)
Himachal Pradesh	5

⁴⁰IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

State	Plazas
Meghalaya	4

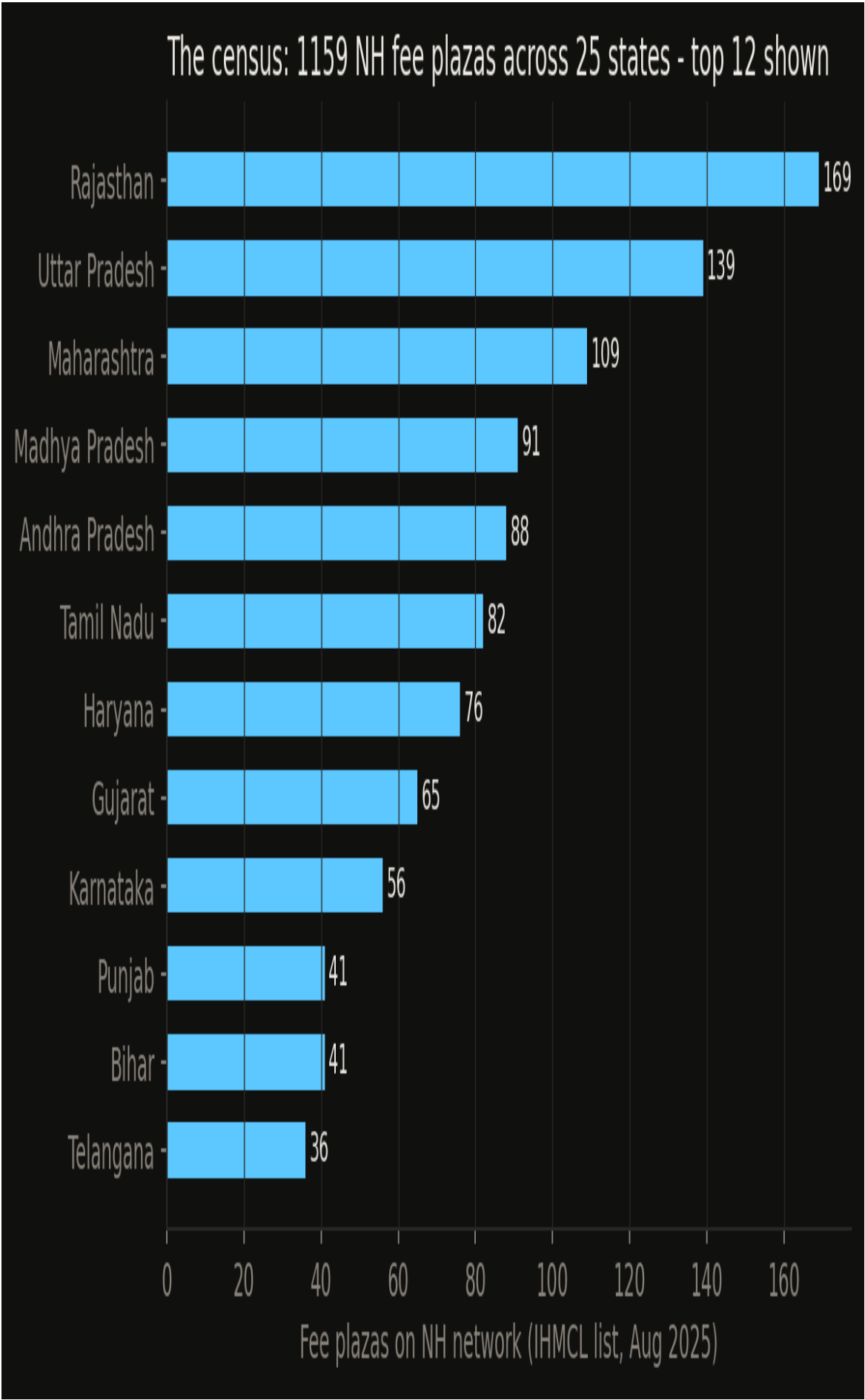


Figure 4: The official registry, counted by state. 1,129 National Highway fee plazas across 22 states/UTs in the IHMCL list of August 2025 – the same registry that still lists Atraila Shiv Gulam today. Source: IHMCL NH Fee Plaza list, Aug 2025 (see Appendix D).

(Parsing note: ~1,133 rows parsed cleanly of a reported ~1,150+; a handful of wrapped rows could not be state-resolved and are excluded from state counts. Raw CSV and script are in the evidence archive.)

The headline: the scam plazas were concentrated in exactly the states where plazas are densest — **UP (135 plazas in the registry) contributed the largest single block of the 42-plaza list (10)** — but the footprint is genuinely national: 14 states in the STF list alone, from Jammu to Assam.

5.1.1 The second open-data source: NHAI's own Toll Information System

The IHMCL list is only names and locations. NHAI's **Toll Information System** (TIS, tis.nhai.gov.in) goes further: it is the ministry's own map-based registry, and its underlying data feed carries, per plaza, the fee-notification date, the commercial operation date, the **concessionaire/operator of record**, traffic versus target figures, and per-vehicle fee schedules — the exact fields an auditor would need to reconcile what a plaza *should* have collected against what it *did* remit.

That feed is now heavily access-controlled, but a full snapshot from August 2022 (782 plazas with structured fields) survives in public web archives and is republished with this report⁴¹. It matters for three reasons. First, it shows what *was* technically feasible years before the scam surfaced: per-plaza operator identity, fee schedules and coordinates in machine-readable form. Second, it records Atraila Shiv Gulam as a **hybrid-annuity-model (HAM) plaza** with an individual operator of record — the structure that let one contractor's booth computers run unauthorised software for years. Third, its fee-schedule fields let anyone reproduce the arithmetic of a skim: at Atraila's recorded car fee of ₹80, the STF's claimed ₹45,000/day of pocketed cash implies roughly **560 unticketed car-equivalents a day** at a single booth.

⁴¹<https://web.archive.org/web/2022/https://tis.nhai.gov.in/TollPlazaService.aspx/GetTollPlazaInfoForMapOnPC>

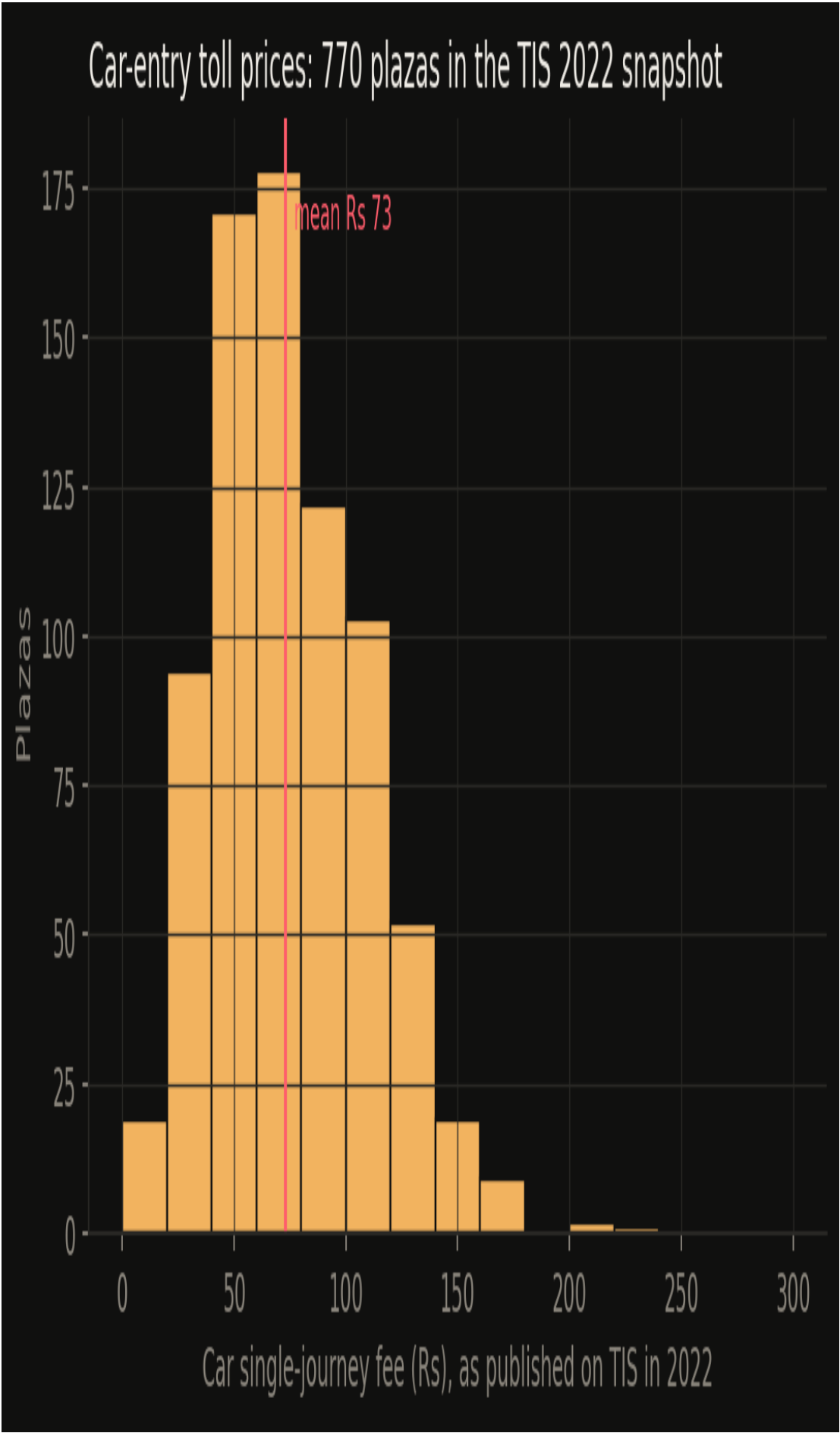


Figure 5: What one car pays, plaza by plaza - distribution of single-journey car fees across 782 plazas in NHAI's August 2022 Toll Information System snapshot. The long right tail (₹300+) is where a cash skim earns the most per unticketed vehicle. Source: NHAI TIS snapshot, Aug 2022 (see Appendix D).

The scandal within the open data is the *decay*: the 2022 feed was richer in audit-relevant fields than anything NHAI publishes openly today. Rebuilding that transparency — operator identity, fee notifications, and reconciliation data per plaza — is a cheaper control than any camera.

5.2 13. Cross-walking the scam list against the registry: Atraila is still listed

We matched the reported scam plazas against the official registry. Three results matter:

1. The scam plazas are ordinary plazas. Atraila Shiv Gulam appears in the registry as *Atralia SivGulam*, NETC code **320132**, district Mirzapur, on the “Varanasi to Hanumana pkg-III” section — no flag, no annotation, indistinguishable from the 1,132 honest plazas⁴². The same for Patni Pratappur (NETC 536051, Shamli district) from the scam list. **There is no public marker in the government’s own data that anything was ever wrong at these plazas.**

2. The registry still lists them. As of the August 2025 edition — after the arrests, after the debarments — the plazas remain listed. We take no position on whether they should have been removed; but a citizen or journalist using only official data would never learn that a scandal had touched these locations.

3. The crosswalk is incomplete — by design, perhaps. The Baghpat scam plaza cannot be uniquely matched to one of the five Baghpat-district plazas in the registry from press names alone. Names in press lists (“Bagpat”, “Chikli”, “Shauly”) do not cleanly match registry names (“Jiwana”, “Baleni”, “Bada-gaom”...). **Matchable only via the concession contract or NETC code — neither public.** An RTI for the 42-plaza list with NETC codes is the obvious next step (see §19).

5.3 13A. The ToT question: are any of the scam plazas institutional-leased assets?

NHAI has been selling 15-to-30-year leases on finished, public-funded toll roads through the toll-operate-transfer (ToT) programme — Macquarie-AIF paid ₹9,681 crore for the first bundle (681 km, ten plazas in Andhra Pradesh and Gujarat, 2018); Cube Highways paid ₹5,011 crore for the third (566 km, nine plazas in UP, Jharkhand, Bihar and Tamil Nadu, 2019-20); later bundles went to IRB (Lalitpur-Sagar-Lakhnadon, ₹4,428 crore), Cube again (Allahabad Bypass, ₹2,156 crore), and others, with ToT awards continuing into 2026⁴³. If a plaza is a ToT asset, its tolling is run by an institution that paid the exchequer upfront for every rupee of fee — an obvious first suspicion for anyone asking where shadow software could hide.

The July 2026 fee-plaza master answers this directly, because it carries a concession-type marker per plaza (TOT / PF / Conc. / STATE). Cross-walking all 42 Jagran-list plazas against the master: **every plaza whose row is legible resolves to PF (public-funded) or a state-highway authority — zero resolve**

⁴²IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁴³ToT programme context and bundle values: domain-b (Bundle 3 award, 20 Oct 2020): <https://www.domain-b.com/economy/infrastructure/roads/nhai-receives-rs5-011-cr-upfront-for-bundle-3-of-toll-highway-projects> · Hindustan Times (Bundle 1 payment, 2018): <https://www.hindustantimes.com/india-news/australian-firm-pays-govt-rs-9-681-cr-to-operate-nine-highway-stretches/story-kagSI9JZSFqpcUb6YifAul.html> · Economic Times (Tot 11/12 awards): <https://economictimes.indiatimes.com/industry/transportation/roadways/cube-irb-top-bidders-for-nhai-tot-13-14-auction/articleshow/106079036.cms> · Indian Infrastructure (ToT status to Jan 2026): <https://indianinfrastructure.com?p=265905>

to TOT. Atraila Shiv Ghulam: PF (its TIS record independently shows project type HAM). Mungari, Harro and Umapur near Prayagraj: PF — they are *not* the Cube-leased Allahabad Bypass. The master's ToT-flagged plazas — the Saurashtra cluster (Morbi/Rajkot/Kutch/Porbandar), the Godavari belt, Jhansi-Lalitpur-Rae Bareilly (matching ToT bundles 11-12), Ranchi, Ghaziabad (Maple Highways), Bihar — share no name with the scam list. Two flagged scam items are *state* roads (MPRDC's Shahdol; Chikliya on a Ratlam state highway), which sit outside NHAI's oversight systems entirely.

Why this matters. The scam concentrated exactly where institutional money is absent: public-funded plazas farmed to low-bid user-fee contractors under two-year cancelable contracts, plus state-highway plazas with no NETC-grade audit at all. ToT concessionaires answer to global credit committees and a 30-year P&L; the shadow app ran where nobody had multi-decade skin in the game. That asymmetry is itself a policy finding: monetisation buys the exchequer upfront and creates one tier of tolling with real oversight, while the un-monetised majority runs on the honour system.

Caveats: a dozen list names could not be resolved in the garbled text layer of the master; matching is by name; and the NHAI InvIT's seed portfolio has not been cross-checked here (an RTI question, drafted). The finding is “no evidence of ToT overlap”, not proof of structural exclusion.

5.4 14. The companion scams nobody counted: receipt tampering, pass misuse, and a driver's complaint at Baghpat

The registry also lets you see what *didn't* get counted. In 2025–26, GST-seat reporting and Hindi press documented a parallel SOP at UP plazas including Itaunja and Nawabganj (Lucknow): trucks running on **~900 fake vehicle-registration numbers per month** to dodge toll entirely — a different exploit of the same manual lane⁴⁴. Consumer complaints continued too: in March 2026, a Baghpat commuter produced a receipt **timestamped 8:46 PM for an 11:30 AM crossing** — the shadow-receipt signature persisting in the wild after the “fixes”⁴⁵. That plaza, ironically, is on the debarred-operator's list.

The lesson for DPI watchers: every digital-public-infrastructure system that delegates the *edge* — the booth, the merchant, the CSC, the banking correspondent — to a private contractor inherits this exact risk: the edge can run unauthorised software, and the centre's audits will read only what the edge chooses to write.

⁴⁴<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁴⁵<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

6 Part V — The state's stake and the driver's data

6.1 15. Why the lane stayed dark: the state was buying data, not control

The exception lane was not an accident that no one priced. It is the residue of a deliberate design choice, made in 2019, about what the toll system was *for*.

When India chose a centralised RFID clearing house over anonymous stored-value instruments (the metro-card model), the argument was never only about queue times. The design — as analysed at the time in *The Wire*⁴⁶ — had three properties worth restating now:

- **A 4 percent digitisation levy on every rupee of toll** — split between issuer bank (₹1.50), acquirer (₹1.25), IHMCL (₹1.00) and NPCI (₹0.25) per ₹100 collected — a recurring fee that contractors recoup from users inside the gazetted rate. The state privatised the upside of collection; the shadow app was the operators' attempt to privatise the cash floor back.
- **Toll receipts as financial collateral.** The same month the FASTag mandate was pushed through, the CCEA cleared toll securitisation and the first toll-operate-transfer (ToT) bundles went to Macquarie (₹9,681 crore) and Cube Highways (₹5,011 crore). The machine-readable toll stream exists substantially so that 15,000+ km of roads can be repriced, bundled and sold against it. The FASTag data is the asset's prospectus.
- **The data itself was the point.** The article's warning — “the ultimate aim... to capture all activities on India's national highways in machine-readable format”, with CCTNS and e-Way Bill integration proposed — reads differently in 2026. What it predicted was not surveillance creep but *audit asymmetry*: enormous institutional investment in what the data can earn, and none in whether the lane event, the keyed record and the remittance are the same thing.

That asymmetry explains the scam's two-year run better than any conspiracy. The data consumer was an investor pricing a ToT bid off NETC flows; nobody's model priced the 8-10 percent of transactions on the booth floor. When NHAI later moved to make FASTag data a monopoly asset and even exempt pieces of the ecosystem from transparency obligations⁴⁷, it confirmed the hierarchy: the aggregate mattered, the receipt did not. The ED's money-laundering case is, in effect, the state finally pricing the receipt.

6.2 16. The privacy dimension: what the system knows — and the data-expansion debate

The same 2019 design analysis set out what a centralised toll system means for the person behind the windshield⁴⁸; a companion piece that month noted FASTag's straightforward use for movement

⁴⁶Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

⁴⁷Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

⁴⁸Srikanth Lakshmanan, “FASTag: Will Datafication of India's Tolls Boost Highway Development?”, *The Wire*, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

tracking in the post-26/11 surveillance architecture⁴⁹. Six years on, the points stand — and the scam’s aftermath sharpens them:

- **The centralised database exists.** Every NETC transaction flows through a clearing house that knows vehicle, issuer bank, plaza, timestamp, and amount — one of the largest vehicle-movement databases in the country, built before India had a data protection law (the DPDP Act arrived only in 2023, and toll systems were never retrofitted with a data-governance review).
- **Anonymity was available and declined.** Anonymous prepaid tags — no bank linkage, no traceability — were technically feasible when the system was designed. The mandate instead required KYC’d, bank-linked tags. The TIS fee schedules even publish *local discounts* (district-registered commercial vehicles pay concessionary monthly pass rates): proof that the system’s economics already encode where you live and what you drive.
- **The blacklist is a geofence waiting for a use case.** The clearing house can update tag blacklists within 15 minutes. The stated use is commercial-vehicle enforcement; the capability is disabling a vehicle’s road access nationally at short notice. In 2019 this was a warning; in 2025, NHAI ordered issuer banks to validate registration numbers and blacklist mismatches — a genuine anti-fraud measure for MLFF, and simultaneously a working prototype for mobility control.
- **The scam is now the justification.** Every post-scam fix — ANPR at every lane, MLFF plate-reading, unified ATMS software, VRN validation — deepens data capture. None of the post-scam procurements we reviewed (§18) contains a consumer data-governance clause: retention limits, purpose limitation, access logs, or an independent audit of who queries the movement database. The fraud is real, and so is the risk that it becomes the standing excuse for a surveillance infrastructure nobody chose.

The consumer-position summary is simple: **you were double-charged in a lane that was never audited, by a system that knows exactly where and when, and the data trail that proves your case is owned by institutions that are not accountable to you.** NPCI, the clearing house, has been explicitly structured outside RTI reach⁵⁰. An RTI to NHAI gets you a debarment order; an RTI to NPCI gets you nothing.

7 Part VI — The fix, and whether it fixes

7.1 17. What changed after January 2025 — and what, two years on, still has not

⁴⁹“After 26/11 Mumbai Attacks, India Has Weaponised Economic Transactions for Mass Surveillance”, The Wire: <https://thewire.in/government/26-11-mumbai-attacks-india-weaponise-economic-transactions-mass-surveillance>

⁵⁰Srikanth Lakshmanan, “FASTag: Will Datafication of India’s Tolls Boost Highway Development?”, The Wire, 14 December 2019: <https://thewire.in/economy/fastag-will-datafication-of-indias-tolls-boost-highway-development> (full archived text in the evidence bundle).

Control	When	What it does	Verdict
AI audit cameras + ANPR pilots	announced Mar 2025 ⁵¹	Independent vehicle counting at lanes	The <i>right</i> control (independent count); deployment status unknown
Toll Remittance Alert System (TRAS)	tendered Apr 2023 ⁵²	Alerts when agency remittances lag collections	Blind to this fraud class by design — it monitors <i>remitted</i> money; skimmed cash was never booked
Cash ban: FASTag/UPI-only lanes	10–11 Apr 2026	Ends the cash vector entirely	Kills the vector prospectively; transfers friction to consumers without tags
Unified ATMS software (RFP May 2026)	2026	One national plaza-software stack; spec contains no manual-lane concept at all ⁵³	Digitises the exception category itself; removes operator discretion
MLFF (barrier-free tolling) acquirer-bank model	pilot RFP Jan 2025 ⁵⁴	No booth, no operator, no cash — ANPR-based charging	Structural fix; per-plaza procurement model will scale slowly
STQC certification scheme for TMS (tamper-evident databases, audit trails)	scheme live ⁵⁵	Certifies the lane software itself	The control that would have <i>detected</i> the shadow app; unclear it was ever procured for plaza TMS

7.2 18. The procurement critique: three official documents that show NHAI knew where the hole was

Three tenders, read side by side, reveal the pattern:

⁵¹Ministerial statements, Lok Sabha, 20–21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

⁵²IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁵³IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁵⁴IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁵⁵<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

- **TRAS (2023)** monitors remittances only — its data feeds are the agency’s own declarations. It is a watchdog fed by the watched.
- **Unified ATMS (2026)** is a 239-page spec with state-of-the-art cloud architecture — and **zero occurrences** of “manual lane”, “cash lane” or “handheld device”. The exception category has been designed away rather than instrumented. Good for fraud, but the residual risk migrates to ANPR mis-reads (fake and duplicate plates — already a documented racket⁵⁶) and to digital-exception fraud.
- **MLFF (2025)** is being procured **one plaza at a time** (“selection of acquirer bank... at Nemili”) — re-procurement as a scaling strategy is how you get a patchwork of vendors and a decade of transition.

And the deepest gap: **no signed, hash-chained, lane-event data integrity requirement** appears in any plaza-software procurement we reviewed. The TMS remains software the contractor buys from whomever it likes, certified (if at all) against no public register. STQC’s certification scheme — tamper-proofing, audit trails — exists on paper; we could not find a public record of a single plaza TMS certified under it. *(RTI filed to confirm.)*

7.3 19. What we still don’t know - and are filing right-to-information requests to find out

Our RTI applications (drafted, ready to file at rtionline.gov.in⁵⁷) target the remaining unknowns:

1. **FIR 0017/2025 with seizure lists** (SP, Mirzapur) — the primary evidence of what the software actually did.
2. **The 14-agency debarment file**: plaza-wise map with NETC codes, PBG ledger, penalty orders (NHA) — the accountability ledger the HC gutted.
3. **TRAS status** (IHMCL): vendor, go-live date, alerts generated to date.
4. **The Atraila concession chain** (NHA RO Varanasi): who “Veltek” legally is, who the concessionaire was.
5. **STQC certification register** for plaza TMS products.

⁵⁶<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁵⁷Workspace archive: Documents/research/FASTagScam/rti/RTI-DRAFTS.md.

8 Part VII — What it means

8.1 20. What it means for you: the driver who pays and the driver who pays twice

If you travelled a non-FASTag vehicle through any of the 42 (now ~100) plazas between roughly 2023 and 2025, you probably **paid double toll in cash and received a receipt that never entered any government system**. Your legal remedy as an individual is effectively nil — the amount is small, the counterparty anonymous, and refunds for all faulty collections nationwide in 2024 totalled ₹12.55 lakh⁵⁸ — against a diversion measured in the hundreds of crores. The real loser is the exchequer; the real payer is you again, through taxes and future toll resets.

What you *can* do:

- **Keep FASTag active and linked** — the manual lane is the risk surface; digital lanes are settled and traceable.
- If you get a suspicious toll receipt (wrong timestamps, wrong plaza, amounts that don't match the schedule), report it to NHAI's helpline (1033) **and** to us — the Baghpat complaint shows the pattern persists.
- Support the boring fixes: independent lane counting, public concession data, certified lane software. That is where this stops.

8.2 21. What it means for digital public infrastructure: the blind spot is a design pattern

This case is a template, and it should be studied beyond highways:

- **The edge is the attack surface.** UPI's QR codes, ONDC's seller apps, AgriStack's field agents — every DPI stack delegates its edge. An unauthorised app at the edge, showing the centre what the centre expects to see, is a *general* attack pattern.
- **Auditing the ledger is not auditing the world.** NHAI audited its TMS. The TMS was fed by the contractor's lane software. The shadow app never touched the TMS. **Reconciliation must anchor to physical events** (vehicle counts, not keyed records).
- **Punishment is not accountability.** NHAI's ₹100 crore crackdown evaporated in court because the evidentiary work hadn't been done. The ED probe is attempt two. In DPI governance, the sequence matters: forensics first, penalties second.
- **The cash-policy dimension.** The official response to date - the double-fee rule, the April 2026 cash ban, the all-digital ATMS spec - narrows the role of cash at plazas. This scam ran on software, not on cash's existence; cash lanes also functioned as the one independent, offline cross-check on the digital record. Whether removing them improves integrity or removes the last external check is a policy judgment the procurement record does not yet answer. Section 23 states our position.

⁵⁸Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of "over 98 percent" per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

8.3 22. Closing the gap: digitisation, audit capability, and residual risk

Every official response to this scam so far points the same direction: less cash, more software. Double fees for cash payers (November 2025). Cash banned at plazas outright (April 2026). A Unified ATMS that consolidates all plaza software into one national platform, and MLFF barriers-free tolling built on camera-based plate recognition. TRAS alerts on remittances. Each step deepens digitisation. None of them audits software.

This case gives that direction a mixed report card:

- **The audit deficit was the vulnerability.** For two years, no IT audit function anywhere in the chain - IHMCL's device certification, NHAI's TMS audits, concessionaire supervision, bank reconciliation - was capable of detecting unauthorised code running on the plaza's own booth computer. The detection that finally worked was a police sting with seized laptops. That is not an audit capability; it is luck with a budget. Nothing announced since - TRAS, Unified ATMS, MLFF - adds independent code-level audit of what runs at the lane. STQC certification covers hardware devices, not the software stacks that have already been shown to be the attack surface.
- **Each new digital layer is new attack surface.** The shadow app showed the pattern: an unauthorised program at the edge, feeding the centre what it expects to see. Now consider the new layers: an ANPR/MLFF system whose plate reads can be spoofed with printed stickers; a Unified ATMS that centralises plaza software into a single national target - one compromise, thousands of lanes; digital notices by mail/SMS that replace the booth receipt with something no driver can verify at the spot; chargeback and exception flows that are pure software artifacts. India is building a taller stack on the same unaudited foundation, and calling the height a fix.
- **Centralisation concentrates risk.** Distributor-model failure (42 to 100 plazas, different agencies, one shared template) was contained by its very distribution. A unified national tolling platform inverts that: it converts every future lane-level fraud into a potential network-level event, and every platform bug into a national outage. Cyber risk is not reduced by removing cash; it is reshaped - and concentrated - by removing redundancy. Cash lanes were an accidental second channel: independent, offline, hard to fake at scale. Deleting them deletes the system's only built-in failover, in exchange for aesthetics.
- **What an actual fix looks like** is boring and software-shaped: hash-attested, version-controlled TMS builds with tamper-evident boot logs on every booth computer; independent code audits (with published SBOMs - software bills of materials) of lane software, not just certified barcode printers; reconciliation anchored to physical events (inductive-loop or ANPR vehicle counts) rather than keyed records; retained lane-level logs the operator cannot edit; and consumer redress that treats the driver's receipt as evidence, not noise. None of these are in the procurement documents reviewed in section 18; the TRAS RFP audits remittances, the ATMS RFP consolidates operations, and both leave the code question open.

The pattern this case documents is not digitisation failing - it is digitisation being measured only by deployment metrics. Each new layer (Unified ATMS, MLFF, ANPR) deepens dependence on software whose integrity nobody in the chain currently audits. Whether the new architecture reduces risk or

concentrates it is, on the procurement record so far, an open question - one this report takes up in the position statement that follows.

8.4 23. The CashlessConsumer position

This section is advocacy, and is labelled as such. The analysis above ends with open questions; this is where we stand on them.

The record in this case shows a gap between how digitisation was presented - as the end of corruption, as operational efficiency delivered - and what the systems on the ground actually were: unmonitored computers running unverified code, reconciled against themselves, for two years. The gap was closed, at every stage, by narrative: press releases described an accountability that the evidence could not yet support, and the Delhi High Court's judgment is the clearest single correction of that gap.

As CashlessConsumer, we take from this case a simple set of demands, applied to every payment system, digital or not:

- **Transparent** - the public should be able to see what software runs at the point of payment, what it costs, and who is accountable when it is wrong. Version registers, audit summaries and penalty orders should be published, not discovered through police raids.
- **Accountable** - enforcement should rest on evidence assembled to a standard that survives scrutiny, not on announcements. A sanction built on a press release fails the public twice: once when it collapses, and again when the underlying fraud goes unpunished.
- **Privacy-friendly** - tolling data reveals where people travel. Every expansion of collection (ANPR, plate databases, account-level reconciliation) needs a stated purpose, retention limit and audit trail. The scam plazas were cameras and computers watching drivers with nobody watching them.
- **Consumer-oriented** - the driver's receipt, complaint and refund right should be treated as evidence infrastructure, not noise. The Baghpat complaint (section 14) did more for detection than any dashboard.
- **With liberty to use cash** - cash is not the corruption vector; it is the one payment channel that is offline, independent of the software under audit, and available to those the digital rail excludes or overcharges. This case is an argument for keeping it, not abolishing it. Systems that demand trust should be audited enough to deserve it.

There is a deeper lesson about sequencing, and it applies to every mandate in this file. FASTag was made compulsory in 2021; cash was taxed toward extinction in November 2025 and banned at plazas in April 2026 — and at every step the mandate preceded the audit. A payment system people cannot opt out of is, by definition, a system whose fraud controls must be proven before the mandate, not after the first scandal. The tolling stack was made mandatory while its audit capability was still theoretical, and the shadow software moved into exactly that gap: a national obligation to pay, enforced over computers nobody was authorised to inspect.

This is the drag that fraud exerts on the tech-optimist project. Every undetected shadow app converts a citizen's healthy scepticism of "cashless" into hardened distrust; every court-set-aside crackdown

converts a regulator's announcement into evidence that enforcement is theatre. The April 2026 cash ban was sold as the fix for a fraud that software committed and audits missed — so the public record now reads as: digital failed, the response was to remove the alternative, and the audit still has not been fixed. That is how corruption and tech-solutionism feed each other. The promise was that digitisation would end corruption; the reality was corruption running on the digitisation; the response was more mandate; and the mandate's costs — the 2× penalties, the lost cash option, the surveillance of every plate and every trip — are paid by people who were already the victims. The state loses revenue and credibility in the same transaction. Both sides pay; the fraud alone comes out ahead.

In the end, that is the balance sheet of the FASTag bypass: drivers paid twice and got receipts no system recorded; the exchequer lost hundreds of crores and then watched its enforcement collapse in court; operators who may be innocent lost contracts and guarantees on the strength of a press note; and the one party that never had a bad day was the fraud itself — two years of business, detection by police sting rather than audit, and beneficiaries whose securities are recoverable only through arbitration. A mandatory system with immature controls does not eliminate corruption; it relocates corruption to where the audits are not, and hands it a captive customer base. Fraud had a field day because the system was compulsory before it was verifiable.

The deeper lesson is about sequence. India now deploys digital public infrastructure as a mandate first and audits it later - FASTag scaled before lane software was certified, cash abolished before remittance reconciliation existed, e-notices planned before a fraud-redress channel. Mandate-first works only when the fraud-prevention work is already done. Done in this order, every scam becomes a referendum on the technology itself: the shadow app did not just steal from the exchequer, it spent public trust. Each undetected plaza made the next "100 per cent digital" claim less believable; each crack-down that collapsed in court made enforcement look performative. That is the drag fraud places on the tech-optimist project - the scandals produced by weak controls become the strongest argument against digitisation, and the state responds by mandating harder rather than auditing better, which deepens the cycle.

The result is a loss distributed onto everyone except the fraudster. Drivers paid twice - once in cash at the booth, once in taxes for the uncollected fee; the exchequer lost revenue and then lost a ₹100-crore enforcement action in court; and both now pay again for a taller, unaudited digital stack. Meanwhile, the people who ran the unofficial lane for two years made crores. Corruption in this design is not an accident the system failed to prevent - it is a cost the system quietly externalised onto its users. Digital public infrastructure earns mandates by being hard to defraud, not by banning the alternative. Until that order is fixed, both the citizen and the state will keep paying for a lane somebody else built to exploit.

Digital payments built to the standard above are worth having, and worth mandating only after they are demonstrably audit-worthy. The lesson of the FASTag bypass is that India has been building them the other way round.

9 Appendix A — Methodology: the registry parse

- **Source:** IHMCL “National Highway (NH) Fee Plazas” list, PDF, August 2025 edition⁵⁹.
- **Extraction:** `pdftotext -layout`, then a Python parser over the row grammar `S . No | NETC code | Name | State | District | Section | NH | PIU | RO`, with next-line fallback for wrapped name cells and a special case for the Atraila row (“Atralia SivGulam”).
- **Result:** 1,133 rows parsed with state attribution; ~9 unresolved (wrapped rows). NETC codes verified by spot-check against known plazas (e.g., 320132 = Atralia SivGulam; 536051 = Patni Pratappur).
- **Crosswalk:** scam-plaza names from the Dainik Jagran 42-plaza list⁶⁰ were matched on normalised name + district. Unique matches: Atraila Shiv Gulam (320132), Patni Pratappur (536051), Madanpur (Assam). Baghpat and several others are ambiguous (multiple registry plazas per district); we do not force matches.
- **Limitations:** the registry names plazas by their fee-schedule names; press lists use local names. Concessionaire/agency attribution per plaza is **not** in the public registry — it requires RTI or concession documents.
- All scripts, the parsed CSV, and the source PDF are in the evidence archive (Appendix C).

10 Appendix B — The 42-plaza list (preserved)

From Dainik Jagran, 23 January 2025⁶¹ — preserved here because the original article is already dead-linking:

Uttar Pradesh (10): Harro (Prayagraj) · Mungari (Prayagraj) · Umapur (Prayagraj) · Andi Lohra (Azamgarh, AK CC) · Bagpat (AK CC) · Faridpur (Bareilly) · Patni Pratappur (Shamli) · Atraila Shiv Gulam (Mirzapur) · Nainsar (Gorakhpur) · Chikli **Madhya Pradesh (6):** Jangabani · Mohtara (AK CC) · Shalibada · Shahdol · Gahra · Chikhlikala (Chhindwara) **Rajasthan (4):** Phulera (Jaipur) · Kadi Shehna (AK CC) · Shahpura · Shaully (operating company reported as “Anuvejan”, ≈ Innovision — debarred) **Gujarat (4):** Mokha (AK CC) · Rohisa (AK CC) · Okha Mandi · Kuchadi **Chhattisgarh (4):** Maharajpur · Mudiapara · Kumhari (Durg, AK CC) · Bhojpuri **Maharashtra (3):** Dashrakhed · Khani Bade · (third per list) **Jharkhand (3):** Nawasari · Turup (AK CC) · Tand Balidha **Punjab (2):** Dhulal · Jigha (AK CC) **Assam (2):** Madanpur (RK Jain Infra — debarred) · Balachera **West Bengal (2):** Gobari · Pashchim Madati **Jammu (1):** Van · **Odisha (1):** Kadali Garh · **Himachal (1):** Sanwara · **Telangana (1):** Jangaon

AK CC = AK Construction (debarred). Counts per Jagran; the outlet’s state tallies differ slightly from other outlets (we preserve the primary list as published).

⁵⁹IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁶⁰<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁶¹<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

11 Appendix C — Evidence archive and negative results

This report is the analysis layer; the archive is the evidence layer. All 23+ press captures (with canonical URLs), five government PDFs (TRAS RFP, Unified ATMS RFP, MLFF RFP, plaza registry, ATMS training deck), full archived copies of the Delhi High Court judgment in *T. Suryanarayana Reddy v. NHAI*, W.P.(C) 10788/2025 (3 July 2026, Indian Kanoon doc 99533796), the September 2025 *Innovision Ltd v. NHAI* interim-stay order (doc 159332680), and our judgment analysis under `legal/`, the parsed registry CSV, RTI drafts and the master dossier live in the workspace under `Documents/research/FASTagScam/`.

Court record (added September 2026): the full text of the lead judgment, *T. Suryanarayana Reddy v. NHAI*, W.P.(C) 10788/2025 & connected matters (Delhi High Court, Justice Sachin Datta, 3 July 2026), is archived at `legal/` (clean text, raw Indian Kanoon capture in Markdown and HTML, plus our paragraph-level analysis). Section 10's analysis cites paragraph numbers from this text. Also archived: the 12 September 2025 interim-stay order in *Innovision Ltd v. NHAI* (doc 159332680) - the first-round judicial pushback that preceded the batch judgment.

What we looked for and did not find (as of 4 September 2026) — published for scientific honesty:

- No app-store, GitHub, code-hosting or certificate-transparency trace of 'Mobdata'/'Any' or plausible variants.
- No public copy of the ED press release (site's Umbraco API confirms 1,995 releases through 2 September 2026; none on this case yet).
- No public copy of FIR 0017/2025 or the charge sheet (RTI route).
- No public record of any plaza TMS certified under STQC's scheme.
- The Lok Sabha answer PDF (USQ 1017) was unreachable from our network; the ministry's reply is quoted via secondary reporting⁶².
- Caution: "MobData" (MobTech, Chinese analytics SDK) is an unrelated namesake. Several LinkedIn profiles named "Alok Kumar Singh" are unrelated persons; do not chase.
- The Delhi High Court judgment (*T. Suryanarayana Reddy v. NHAI*, W.P.(C) 10788/2025 & connected matters, 3 July 2026, Justice Sachin Datta) — archived full text in `legal/` (lead judgment: Indian Kanoon doc 99533796; interim-stay order in the connected *Innovision* matter: doc 159332680). Analyzed in section 10.

⁶²Ministerial statements, Lok Sabha, 20-21 Mar 2025; FASTag penetration of "over 98 percent" per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

12 Appendix D — Sources

ED/PMLA (Sep 2026): The Tribune⁶³ · Daily Excelsior⁶⁴ · Times of India⁶⁵ · Mathrubhumi⁶⁶ · Bizz Buzz⁶⁷
STF case (Jan–Feb 2025): Times of India⁶⁸ · Aaj Tak⁶⁹ · Dainik Bhaskar⁷⁰ · Dainik Jagran^{71,72} · Navbharat Times⁷³ · Amar Ujala⁷⁴ · OpIndia⁷⁵ · Hindustan Times (Gorakhpur)⁷⁶ **Official response (Mar 2025):** Lok Sabha written answers⁷⁷ · Rediff/PTI⁷⁸ · UNI⁷⁹ · Daily Jagran⁸⁰ · Upstox/PTI analysis⁸¹ **Legal:** Delhi High Court, *T. Suryanarayana Reddy v. NHAI*, W.P.(C) 10788/2025 & connected matters, 3 July 2026 (primary; Indian Kanoon docs 99533796, 159332680)⁸²; India Legal Live (Delhi HC, 6 Jul 2026)⁸³ **Policy & tech documents:** IHMCL TRAS RFP⁸⁴ · Unified ATMS RFP⁸⁵ · MLFF Nemili RFP⁸⁶ · IHMCL plaza registry⁸⁷ · STQC TMS certification scheme⁸⁸ **Consumer corroboration:** TV9 Hindi (Baghpat, 5 Mar 2026)⁸⁹ **RTI drafts:** workspace archive⁹⁰

⁶³The Tribune, 4 Sep 2026: <https://web.archive.org/web/20260904110010/https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/> (live URL intermittently blocks automated access)

⁶⁴<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

⁶⁵<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

⁶⁶<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

⁶⁷<https://www.bizzbuzz.news/crime/ed-raids-14-locations-across-india-in-toll-plaza-scam-seizes-over-rs-1-crore-cash-1403756>

⁶⁸<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁶⁹<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁷⁰Dainik Bhaskar, 23 Jan 2025. Original link (<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>) is now dead; full archived copy in the public evidence bundle: <https://zo.pub/cashlessconsumer/fastagscam-open-data>

⁷¹<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁷²<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁷³<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

⁷⁴<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

⁷⁵OpIndia, 25 Jan 2025: <https://web.archive.org/web/20250720160857/https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁷⁶Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

⁷⁷Ministerial statements, Lok Sabha, 20–21 Mar 2025; FASTag penetration of “over 98 percent” per PIB, 11 Jul 2025: <https://pib.gov.in/PressReleasePage.aspx?PRID=2143962>

⁷⁸<https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

⁷⁹<http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

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⁸¹<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁸²<https://indiankanoon.org/doc/99533796/> (lead matter, W.P.(C) 10788/2025, decided 03-07-2026). Full archived copies in Documents/research/FASTagScam/legal/.

⁸³<https://indialegallive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

⁸⁴IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁸⁵IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁸⁶IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁸⁷IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁸⁸<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

⁸⁹<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

⁹⁰Workspace archive: Documents/research/FASTagScam/rTI/RTI-DRAFTS.md.

Report by CashlessConsumer research (agentic workflow, human-supervised). Feedback: cashlessconsumer@zo.computer. This report will be updated when the ED press release and RTI responses land. Companion page: cyber.cashlessconsumer.in.

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